

Faculty of Computer Technology and Cybersecurity
Department of Computer Engineering
ARCHITECTURE AND ORGANIZATION OF COMPUTER SYSTEMS
WORKSHEET - 1

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Q1: Describe Computer Architecture

A computer architecture is the conceptual design of a computer system from the programmer's view. It describes what the computer does and what is visible to software: instruction set (ISA), data types, addressing modes, input/output operations and system behavior.

Q2: Explain the structure and role of the processor and its components

The processor (a.k.a CPU), executes program instructions and controls the work of the computer system.

The main components:

- **Control Unit (CU):** fetches and decodes instructions, generates control signals, and coordinates other computer components
- **Arithmetic Logic Unit (ALU):** performs arithmetic operations such as addition and subtraction, and logical operations such as AND, OR, and NOT.
- **Registers:** very fast storage locations. They hold operands, addresses, instructions, and intermediate results.
- **Cache memory:** small and fast memory that stores frequently used data and instructions.
- **Memory interface:** manages reading data from and writing data to main memory.
- **I/O interface:** lets the CPU communicate with external devices such as a keyboard, disk, or network.

Q3: What are some technological developments in Computer Architecture?

1. Vacuum tubes: They were used in the first electronic computer. They were large, slow, unreliable, and consumed much power.
2. Transistors: They replaced vacuum tubes. They made computers smaller, faster, more reliable and produced less heat.
3. Integrated Circuits (IC): This is a piece of silicon with many transistors on it. It increased speed, reduced cost, and enabled mass production.
4. Microprocessors: Putting an entire central processing unit (CPU) onto a single integrated circuit made computers incredibly small, fast, and affordable enough for personal use.

5. Advanced technologies: modern systems use multi-core CPUs, nanotechnology, GPUs, TPUs, AI accelerators, and specialized processors.

Q4: What is the difference between Computer Architecture and Computer Organization?

Computer architecture defines what commands the CPU understands and how programs can use them: the instruction set (ISA), data types, registers, and memory addressing. Computer organization defines how the hardware is built inside to execute those commands: CPU parts, cache, buses, and control signals.

Q5: Explain the different types of data manipulation

Data manipulation is about processing data to get useful information:

- **Arithmetic:** add, subtract, multiply, divide
- **Logical:** AND, OR, NOT
- **Comparison:** check whether values are equal, greater, or smaller
- **Conversion:** change data from one format to another, for example decimal to binary